The CIRIS Constitution: The Reader's Policies

The Reader's Policies, a volume of edition 1.0-rc12

Stewarded by Eric Moore

The Reader's Policies

What this is. How a consumer decides whom to trust: the anti-patterns that a composition must not fall into (4.1), and the composition and delegation policies of 4.4, Policies A to M and the aggregation rules, with the delegation graph's depth cap, cycle rejection and weight cap.

Whom it binds. Every CEG-Conforming Consumer that composes rows into a verdict, and every reader who wants to know what a score about them will be turned into. A consumer may add policies of its own; it may not skip the ones marked normative.

What it promises. That a verdict is a function of signed rows and declared policy, reproducible by anyone holding the same rows; that one human voice counts once (1.13.2, with the effective count of 6.1.2.1.2); that delegation is scoped, attenuated and revocable per link, that a hierarchy delegation ends within a year (4.4.3.2.8) and an act-on-behalf delegation ends at its stated `delegation_valid_until`; that a grant confers no re-grant and a licence no issuance, and that only a `grant`-scoped delegation from the owner confers issuance, within the depth cap and the yearly term.

What it leaves out. The rows themselves, which are *The Wire* and *The Dictionary*; the authorities whose rows a reader chooses to trust, which is *The Constitution*; and the mathematics of coherence under which the aggregation is analysed, which is *The Mathematics*.

Where the rest lives. Each policy's section states its inputs and its output in prose.

Part 4 — Composition & Governance

4.1 Anti-patterns

These are wire-format reaches that fail the CC 1.2 operational-language gate or import Cartesian-individualist defaults. They are recorded so that an author finds the discipline before reaching for the wire format. A CEG-Conforming Producer (CCP) SHOULD NOT emit attestations matching the anti-patterns below.

4.1.1 Delegation-laundering anti-pattern

`delegates_to` → `delegates_to` → `delegates_to` →... → `attacker` chains, where each hop is individually well-formed but the aggregate routes trust to a terminal attester the original delegator would not have approved.

What's wrong	Correct expression
Unbounded depth `delegates_to` chains	Consumer policy MUST cap traversal depth at **5 hops** by default (configurable); chains longer than the cap are treated as `attestation:self_verify` only (no transitive trust). A substrate MAY additionally clamp any caller-requested depth at an absolute ceiling; the ceiling bounds the walk, it does not raise the default
Cycles (A → B → A)	Substrate MUST detect cycles on the `delegates_to` graph and reject the cycle-closing emission
A self-edge, or a trust-plane edge, refused as a cycle	A cycle needs **two or more distinct keys**: `delegates_to(A → A)` is the charter form (CC 3.2 T1 self-root; `trust:charter:v1` is self-referential by definition, CC 3.4.7), not a cycle. The graph the row above is over is the **capability** graph: a `trust:accepts:v1` acceptance edge (CC 3.2 T3, node → root) is the node's un-trust lever and confers nothing on the root, so it is outside the cycle check in either arrival order; a capability `delegates_to` closing the same pair is still refused
Aggregate-weight concentration	Consumer policy SHOULD cap the trust weight any single terminal delegate can accumulate from a given root attester at **0.5** × **root_trust** by default

4.1.2 Discipline pattern

The recurring shape across most anti-patterns: **extending the wire format so single attesters can pre-declare their own state more richly**. Each one is reachable, none is necessary.

The Ubuntu-primary discipline cuts cleanly: standing is constituted relationally through attestation by others, not through self-declaration. The wire format should **resist** primitives that let a single key announce its own state without external composition. The substrate is austere by design so consumers compose verdicts; richer narrative expression belongs in `context:`, `evidence_refs[]`, and downstream witness attestations, not in new envelope enum members or new self-attestation prefix families.

4.1.3 Rejected wire additions

Anti-pattern	What it would smuggle	Correct expression
`detection:emergent_deception:{axis}`	Moral verdict ("deception") in prefix name	`detection:correlated_action:{axis}` — mechanism-descriptive
`attestation:l{N}:*`	Ladder-position (verdict-shape) in wire prefix — same shape as `score:trustworthiness:*` smuggling meta-judgment as wire	`attestation:{mechanism}` bare mechanism (`self_verify` / `hardware_rooted` / `registry_consensus` / `license_validity` / `agent_integrity`); consumer composes L1-L5 ladder via CC 4.4.3.6 Policy I
`score:trustworthiness:{entity}`	Meta-judgment as separate prefix	Compose downstream from `licensure:*` / `capacity:*` / `provenance:*` attestations
`flag:bad_actor:{axis}`	Pejorative wire vocabulary	Surface as low-confidence scores on `provenance:*` and `coherence_standing:*`; adjudicate by WA quorum (CC 4.5.1)
`grounding:{tradition}:{principle}`	"Tradition" claims are interpretive, not mechanism	Reuse `delegates_to` structural primitive per CC 2.4.1.2
`infra:observe` (or any `observer:*` capability)	Observation as a *granted* capability — inverting consent, so a token the observer carries would decide what may be seen	**No scope, deliberately.** Observation is the zero state: "we begin as an observer" is the base self-attestation existing at all (CC 4.4.3.8). What may be observed is governed by the subject's consent — `cohort_scope`, `key_grant`, observer-share — never by a capability the observer holds.

4.1.4 `withdraws` arbitrage

Per CC 2.4.1.3: a misattester can `withdraws` instead of `recants` to dodge the trust penalty consumers apply to acknowledged-error chains.

What's wrong	Mitigation
Asymmetric attester incentive	Consumer policy MUST track per-attester `withdraws:recants` ratio over a rolling window; attesters whose ratio exceeds a configured threshold (default 5:1) SHOULD be downweighted regardless of which structural primitive they use. The `recants` distinction matters per CC 2.4.1.3, but the practical anti-arbitrage countermeasure is consumer-policy behavioral analysis, not a wire-format change.

4.1.5 Rejected stress-test reaches

Each entry below reaches for a richer self-declaration; each is reducible to existing primitives.

Anti-pattern	Why reject	Correct expression
`epistemic_mode: introspection`	Cartesian shortcut — lone subject pre-declaring inner state as if that constitutes standing	`witness_relation: self` + `confidence < 1.0` + pending external composition. The wire makes introspection awkward to express, because the framework's claim is that self-knowledge does not constitute standing without external witness.
`epistemic_mode: testimony`	Reducible to existing values	`epistemic_mode: external` + `witness_relation: external`
`transparency:{kind}` standalone prefix	Disclosure is constituted by reception, not announcement. Cartesian self-claim shape.	`evidence_refs[]` carries the reasoning-chain hash; downstream `transparency_log: inclusion` from a witness who actually retrieved it.
`stake: civic` / `epistemic` / `dignitary`	`civic` = `stake: reputational` + `cohort_scope: community`. `epistemic` = `confidence` + `stake: reputational` (same axis as confidence; not separate). `dignitary` lives on wrong axis (stake names what the attester loses; dignity harm is what the attested loses → belongs in `harm_class:dignity_harm`).	Compose existing values with cohort/harm-class.
`oversight_mode: deferred` / `active` / `advisory`	All map to existing HITL/HOTL/HOOTL	`deferred` = HITL pre-decision; `active` = HITL with substrate monitoring; `advisory` = HOTL
`provenance_walk` as wire primitive	UX concern smuggled into wire format	Consumer-side composition (the operator's dashboard, agent introspection); the chain already walks via `references_attestation_id` + `topical_relation:*` + `valid_until`
Renaming canonical capacity factors and HE-300 benchmark categories (CC 8.8.10) to "kid-friendly" names	Canonical names map to a worked-out epistemic/ethical lattice that loses precision under accessibility renames	A translation glossary (spec name ↔ narrative name) + version pinning in worked examples

4.4 Composition policies

The substrate carries edges (attestations); consumers compose traversals (verdicts). CEG specifies a library of named reference policies. A CEG-Conforming Consumer MUST implement at least Policy A; the others are RECOMMENDED for richer compositions.

4.4.1 Frickerian discipline: consumer-policy norms

Consumers SHOULD apply identity-prejudice-resistant weighting. Concretely:

- Don't downweight `testimonial_witness:*` from cohorts with low overall attestation density (testimonial preservation is precisely what corrects for that low density).
- Don't downweight `non_maleficence:*` claims about a partner just because the partner has a long `partner_role:*` track record (the long track record may be the harm).

- Apply CC 4.4.3.9 lexical-vulnerability-priority in tie-breaks involving small cohorts.

Adversarial caveat: the discipline above is consumer-policy-only; an adversary can emit `testimonial_witness:victim_of_my_competitor` exploiting the Frickerian non-downweighting rule. Per CC 3.1.9.3, `testimonial_witness:*` is never sole evidence for `slashing:*`; per CC 3.4.7 the consumer MUST also weight `witness_relation: self` claims against the attester's other-emission track record. The Frickerian rule applies AFTER these structural safeguards, not before them.

4.4.2 Aggregation semantics: opinionated defaults

Per dimension+attested_key_id, the verdict is computed by composing attestations under the chosen policy. Default aggregation by polarity column (CC 3.1):

Polarity column value	**Default aggregation**
`signed`	**Mean** of `score` × `confidence` across attesters
`boolean-via-score`	**Min** (any negative trumps positive — fail-secure for hard constraints like `prohibited:*`)
`+1.0 only` / `positive-only`	**Max** across attesters (any positive is conclusive)
`-1.0 only`	**Min** across attesters (any negative is conclusive)
`enumerated`	**Most-recent** by `signed_at` from the attester(s) authorized to emit per CC 3.4
Detector dimensions (`detection:correlated_action:*`, `detection:distributive:access:*`, `ratchet:flag:*`)	**Median** across attesters (resists adversarial mean-pulling by a single captured detector)

Specific dimensions override via consumer policy; the defaults above are the CC 2.2 CEG-Conforming Consumer (CCC) minimum.

4.4.3 Reference policies

The named reference policies below give consumers a shared library for turning edges into verdicts. Policy A (direct trust) is the conformance floor; the rest are RECOMMENDED for richer compositions and are referenced freely across this Part. The sections run in composition order, not letter order; the index by letter:

Policy	**Section**	**Name**
A	CC 4.4.3.8	direct trust
B	CC 4.4.3.12	one-hop transitive
C	CC 4.4.3.13	weighted graph
D	CC 4.4.3.9	lexical-vulnerability priority
E	CC 4.4.3.1	locality-scaled quorum
F	CC 4.4.3.7	`agent_files` trust composition
G	CC 4.4.3.11	trust-fresh / lighthouse
H	CC 4.4.3.3	tiered-scope composition
I	CC 4.4.3.6	attestation-ladder composition
J	CC 4.4.3.10	trusted-publisher composition
K	CC 4.4.3.5	consent composition
L	CC 4.4.3.4	self/family membership composition

Policy	Section	Name
M	CC 4.4.3.2	community membership composition

4.4.3.1 Policy E: Locality-scaled quorum

WA quorum size is a function of decision locality, because a narrow cell cannot seat a fresh recused quorum:

```
quorum_size(scale) = f(locality:decision:{scale})

reference function (policy-tunable):
  local -> 2
  regional -> 3
  national -> 4
  federation -> 6

minimum cell-pool requirement for fresh-quorum recusal at scale S:
  min_pool(S) = quorum_size(S) x 2
```

Recusal is feasible when `cell_pool` $\geq$ `min_pool(S)`. Decision-scale-matching is structurally enforced; overreach surfaces as a named "locality mismatch" failure.

4.4.3.1.1 Sub-quorum fallback

When `cell_pool < min_pool(S)`, the consumer MUST take one of these explicit paths — there is no implicit fallback:

1. **Scale-down**: re-attest the decision at the next-lower `locality:decision:{scale}` (where the smaller quorum requirement is met) AND emit `hard_case:locality_scale_down` so the scaling event is observable for downstream review.
2. **Escalate**: emit `hard_case:locality_underpopulated` and route the decision to the federation-scale cell (which by definition has the largest pool).
3. **Liveness-defer**: emit `hard_case:locality_quorum_unreachable` and defer the decision until the cell pool grows. The deferred state MUST itself be reviewable via subsequent reconsideration.

Recursion safety: the CC 4.5.1 amendment process routes through `locality:decision:federation` by default; the federation cell's pool is sized to make `cell_pool` $\geq$ `min_pool(federation)` always true at federation genesis. If federation-scale pool ever falls below `min_pool(federation)`, the entire amendment surface is in a constitutional crisis state and only the HUMANITY_ACCORD CONSTITUTIONAL halt (CC 4.2) can resolve it.

4.4.3.2 Policy M: Community membership composition

Per CC 3.2 `community` + CC 3.3.3 `location_proof`. Composition pattern for resolving the **current membership set** of a community, gating cohort-filtered visibility for `cohort_scope: community` content.

Sibling to CC 4.4.3.4 Policy L (self/family) but with different defaults — community content is encrypted under a per-community DEK and its holding claims reach members only; the privacy property is byte-confidential-to-members and existence-invisible-to-non-members as well.

4.4.3.2.1 The three crypto tiers + the Community DEK cascade (normative)

The line is drawn at **"does it have a bounded membership roster?"** — yes → encrypt, no → plaintext. This gives a persecuted community a cryptographic home (a bounded **Community**) distinct from the unbounded **Commons**:

Tier	`cohort_scope`	At-rest	Wire discovery	Reader
self / family	`self`, `family`	encrypted, per-write DEK	**none** (CC 5.2 structural invisibility)	occurrences / family members
Community	`community`, `affiliations`	**encrypted under the community DEK**	**none outside the roster**: `custody:ack:v1` within the cohort (CC 3.1.3.3)	community members (DEK cascade)
Commons	`species`, `biosphere`, `federation`	**plaintext**	`custody:ack:v1` at federation scope — the holder directory	anyone
Public room (`cohort_subkind: public`, CC 3.2)	`community`	**plaintext** (Commons tier, no DEK)	`custody:ack:v1` at federation scope; a `community:listing:v1` row	anyone who joins (open admission); replicated to the roster's nodes

Wire discovery: none for `self` / `family` means none **outside** the cohort. Within it, a device's custody report for a blob (`custody:ack:v1`, `here` | `none`, CC 3.1.3.3) is permitted, at the blob's own scope and replicated only to members — never to a non-member.

Community DEK cascade (MANDATORY). Community content (`cohort_scope: community | affiliations`) is encrypted at rest under a **per-community DEK**; its holding claims are placed at the community's scope and reach its members only (CC 3.1.3.3), and there is no non-member holder of community bytes (CC 6.1.5.3). The community DEK **is the CC 5.1 epoch-DEK cascade applied to `cohort_scope: community`** — *a community is a stream its members subscribe to, cryptographically*: **one** DEK shared across emissions (per-emission cost O(1), not O(members)), wrapped to every current member occurrence at each sealed write (a member admitted since the epoch's DEK was minted is wrapped at the next sealed write in that epoch), the epoch rotated before the next seal after a removal (CC 4.4.3.2.2), **`wrap_algorithm: v2` (hybrid PQC) MANDATORY** (same harvest-now-decrypt-later reasoning as CC 4.4.3.4.1 / CC 5.1). This is **mandatory, not opt-in** — the tier name *is* the guarantee; a persecuted community is protected by *being a community*, not by remembering a flag. (Self and family content is sealed the same way under a fresh per-write DEK (CC 4.4.3.4.1); there structural invisibility is the first boundary and the seal is defence in depth, while a community *federates*, so its DEK is its **sole** confidentiality boundary.) **What the seal covers (normative).** The sealed thing is the content body — an inline body, a chunk, a stream segment. Attestation rows (the envelope, the dimension, the scores, the key ids, the scope, the instants, the payload of a `scores` or composer row), the directory, holding claims, the `key_grant` rows' own metadata, blob provenance and the transparency logs are not encrypted at rest; a node that holds them reads them. The only plaintext bodies at community scope are an authorized `infrastructure` community's and a `public` room's, decided by the community record and never by the writer; Commons scopes are plaintext by tier.

Holder-inspectability principle (normative). A node never holds bytes it does not know what they are: either the holder **inspects the bytes** (Commons — plaintext, maximally inspectable, hence the *preferred* social-distribution mechanism) **or** it is a **member of the cohort** the bytes belong to and holds ciphertext its owner can open (CC 6.1.5.3: a node is in a cohort's audience only through its owner's membership). There is no third case. **Nothing above local**

tier is ever a forced, unattributable opaque blob, and nothing is ever an opaque blob held by a stranger. This is *why* the split is shaped this way and what the eviction rules enforce.

The Commons-tier subkinds (normative). Two `cohort_subkind` values **opt OUT of the mandatory DEK cascade and are Commons-tier (plaintext, holding claims at federation scope, no DEK)**: `infrastructure` (CC 3.2) — `ciris-canonical` and any governance/trust root — and `public` (CC 3.2) — a room anyone may join, which therefore has no confidentiality claim against the public. The trust root cannot be an opaque blob; its entire purpose is public auditability — transparency-seeking, not privacy-seeking. **Node→canonical traces** (conformance / `registry_consensus` emissions to a governance community) are therefore `cohort_scope: federation` (Commons/plaintext, world-readable) — a node enrolling in governance is thereby told its conformance traces are public.

4.4.3.2.2 Forward secrecy on community member removal

With a community DEK the removed member holds the current epoch key, so the rule at CC 4.4.3.4.5 is carried by rotation here: after a removal each minter **rotates its epoch DEK before its next sealed write**, so the removed member receives no new community content. What they already received during membership stays with them (once shared, always shared; no PCS), and nothing is re-encrypted.

4.4.3.2.3 Community admission per `consensus_protocol` + `cohort_subkind`

A proposed membership change is a **new version of the roster record** (CC 3.2 T6), admitted under the group's quorum and chained by `prev_head_digest`, covering the roster rows (widenings, revocations, resignations) that ride their own planes — never a single-signer `supersedes` Contribution, which a rule-1 or rule-2 `withdraws` could retire without the quorum. Substrate evaluates the CURRENT community's `consensus_protocol` (same six canonical kinds as family) AND any subkind-specific admission requirements:

```
admit_community_change(C, proposed: community_record):
  current = resolve_community(C, now)
  protocol = current_community_record(C).consensus_protocol
  subkind = current_community_record(C).cohort_subkind

  signatures_ok = evaluate_consensus_protocol(protocol, current, proposed.signatures)
  if not signatures_ok:
    emit hard_case:community_consensus_protocol_violation:{C}
    reject

  // CC 3.1.3.2: a supersede NEVER adds (membership_supersede_cannot_add);
  // growth rides the widening plane, where every NEW member K needs K's own acceptance
  // of an undeclined, unwithdrawn proposal for C, answered and grown before its expires_at
  for each NEW member K in proposed.members (not in current):
    if not acceptance_admits(K, C, proposed.role_of(K), proposed.signed_at):
      emit hard_case:community_consensus_protocol_violation:{C}  // with the named membership_* rule
      reject

  subkind_ok = evaluate_subkind_admission(subkind, current, proposed)
  if not subkind_ok:
    // For geographic: subkind admission failed (e.g., new member's location_proof
    // not contained in geographic_constraint, or expired location_proof)
    emit hard_case:community_consensus_protocol_violation:{C} // same observability prefix
    reject

  admit
  emit hard_case:community_membership_change:{C}
```

The `hard_case:community_consensus_protocol_violation:{C}` row is the record the substrate emits on a refusal; the refusal itself is returned to the submitter as the token `federation_community_consensus_protocol_violation` (CC 4.5.13). The two are different objects: one is emitted, the other is returned.

The `evaluate_subkind_admission` predicate dispatches per `cohort_subkind`:

```
evaluate_subkind_admission(subkind, current, proposed):
  match subkind:
    "geographic":
      # Per the CC 3.2 geographic admission requirement
      for each NEW member in proposed.members (not in current):
        their_proof = latest valid location_proof from new_member.key_id
        if their_proof IS NULL:
          return false // no location_proof on file
        if their_proof.cell_id NOT contained in current.geographic_constraint.cell_id:
          return false // location outside community's geographic bound
        if their_proof.valid_until passed:
          return false // expired
      return true
    "public":
      # CC 3.2 `public`: open admission -- a standing founder proposal and the joiner's
      # own acceptance; the widening still passes consensus_protocol
      if current.admission == "open":
        for each NEW member in proposed.members (not in current):
          if no membership:acceptance:v1 from new_member naming the standing proposal:
            return false // nobody joins without their own consent (CC 3.1.3.2)
      return true
    _:
      return true // unknown subkinds admit on consensus_protocol alone
```

Operator vocabularies extending `cohort_subkind` provide their own `evaluate_subkind_admission` predicates per the `custom:{id}` consensus_protocol hook pattern from CC 4.4.3.4.2.

4.4.3.2.4 Community membership resolution

For community `C`, the current member set is computed as:

```
resolve_community(C, now):
  held = the group record this node holds as head for C
           // one held version per group id: a new version is admitted only if its signed
           // prev_head_digest names the held head; a later signed instant that does not
           // name it is refused, not preferred (CC 3.2 T6)
  head = held, or the witnessed head where C's charter declares a witness_quorum
           // the witnessed head may be an ancestor of the record this node stores
  roster = fold(head, membership rows effective at now)
           // revocations, widenings, resignations, each admitted per CURRENT consensus_protocol; see CC
     4.4.3.2.3
  return {m.key_id for m in roster}
```

Same shape as `resolve_family` (CC 4.4.3.4.4). Each `member.key_id` is an identity.

4.4.3.2.4.1 Deterministic resolution + member→address resolution (NORMATIVE)

In a CEG/RET stack member resolution replaces DNS+IP: it is a chain of *signed* bindings, and every implementation MUST resolve **identically** (an interop requirement). Two normative pins:

(a) Determinism of `resolve_community`. Where the CC 4.4.3.2.4 computation has choices, they are fixed:

- **`now` semantics** — the resolving Consumer's own clock; membership is evaluated as of `now` (a member is included iff joined ≤ `now` and not removed ≤ `now`). No global clock assumed.
- **"the head"** — the one version of the group record this node holds for the group. A new version is admitted only if its signed `prev_head_digest` names the held head; a version with a later signed instant that does not name it is **refused, not preferred**. Where the group's charter declares a `witness_quorum`, the head is the **witnessed** head (CC 3.2 T6), which may be an ancestor of the record the node stores. The roster is the head **folded** with the membership rows effective at `now` — revocations, widenings, resignations; the fold's instant selects nothing.
- **member ordering** — the returned set is canonically ordered by `key_id` (lowercase-hex byte order) for any downstream hashing/iteration.
- **founder-subset eval** — for `cohort_subkind: infrastructure` (CC 3.2) admission is `evaluate_consensus_protocol` over `{m: m.role == founder}`, NOT all members.

(b) Member → reachable address (the DNS-free resolution). Resolving a member identity to a Reticulum destination:

```
resolve_member_transport(C, now):
  members = resolve_community(C, now) // (a) above; admission verified under the current
     consensus_protocol (founder subset for infrastructure)
  out = []
  for M in members (canonical key_id order):
    occ = the current version of M's identity_occurrence (CC 3.3.6 version rule)
      with transport_destination present, valid at now,
      hybrid-sig verified against M (or an occurrence of M) // CC 3.3.6.2
    if occ is null: continue // no authenticated binding -> skip (fail-secure)
    td = occ.transport_destination
    REQUIRE td.destination_hash == rns_destination_hash( // CC 3.3.6.2.1 pinned two-stage
      td.reticulum_x25519_pubkey, td.reticulum_ed25519_pubkey,
      td.app_name, td.aspects) // NOT a flat-concat SHA-256
    out.push((M, td.destination_hash))
  return out // -> Reticulum announce/path-request per dest
```

The three resolutions and their trust roots: **WHO** = `resolve_community` (the roster head, admitted under the current `consensus_protocol`) · **BINDING** = `transport_destination` (federation-key-signed `identity_occurrence`, CC 3.3.6.2) · **WHERE** = Reticulum announce/path-request (mesh, no CEG). Reachability is never trust (CC 5.3.3.4): a path that answers is not an authorization; only the signed binding + quorum are.

Cold-start (bootstrap). A node that knows only `community_key_id` needs two out-of-band pins: (1) the `community_key_id` itself (trust anchor) and (2) ≥1 **seed `destination_hash`** (reachability) — and, before it attaches, a third thing that is neither: (3) a **witnessed lineage head** of that community inside its charter's `attach_window` (CC 3.2 T4a, T6). Freshness gates attaching, never staying attached. It reaches any one seed, pulls the group record at its head, runs `resolve_member_transport`, verifies founder-quorum against the pinned `community_key_id`, then floats on the live set thereafter. A malicious seed cannot forge membership (quorum signature check) — it can only deny service (try another seed). **Bootstrap reachability ≠ bootstrap trust.** `ciris-canonical` (CC 3.2) is the root community whose seeds clients pin.

4.4.3.2.5 Geographic-community admission flow (worked example)

```
1. Alice publishes a location_proof:
  subject_kind: location_proof
  subject_key_id: alice_root_key
  cell_id: "872830828ffffff" // H3 res 7, ~5 km^2, a city centre
```

```
  cell_resolution: 7
  asserted_at: now
  valid_until: now + 30 days
  cohort_scope: federation // public; the disclosure IS the opt-in

  Substrate admits (resolution 7 <= 7 OK). Emission federates.

2. Alice joins the city community -- three rows, never a supersede
  (a supersede never adds a member, CC 3.1.3.2):
  - an active founder of city-community emits a membership:proposal to
    alice_root_key (role: member, expires_at set)
  - Alice answers with her own membership:acceptance naming that proposal --
    no one is seated without the joiner's acceptance
  - a membership:widening seating {key_id: alice_root_key, joined_at: now,
    role: member} is cosigned under the community's consensus_protocol:
    signatures: [...current_member_sigs] // satisfies majority rule

3. Substrate admission gate (per CC 4.4.3.2.3):
  - signatures_ok: majority rule satisfied [ok]
  - acceptance_admits(alice_root_key, city-community, member, signed_at) [ok]
  - subkind: "geographic" -> evaluate_subkind_admission:
    - alice_root_key's latest valid location_proof exists
    - cell_id "872830828ffffff" CONTAINED in "85283083fffffff" (the community's cell, res 5)
      via CC 2.6.6.2 containment (res 7 >= res 5; parent-walk reaches the constraint)
    - valid_until in the future
  - subkind_ok [ok]
  - admit + emit hard_case:community_membership_change:city-community

4. Alice now receives cohort-filtered visibility for cohort_scope: community content
  with community_id: city-community. At the next sealed write in the community's epoch
  the sealer wraps the epoch DEK to Alice's occurrences (key_grant:epoch:v1, wrap v2);
  holding claims stay within the community's roster. (Only an authorized infrastructure
  community or a public room is plaintext, by its record; this one is neither.)
```

4.4.3.2.6 Delivery extension: `delivery_mode` × Policy M

Policy M's community-membership composition extends to govern the **delivery axis**. The subscriber-set for any push-delivery flow IS a `community` Contribution; "subscribe = join the community"; it inherits revocation, consensus, and structural-invisibility from Policy M unchanged.

Subscriber-set composition:

```
For a Contribution C with delivery_mode = push AND cohort_scope = community:
  subscriber_set(C) = resolve_community(C.community_id, now)
    per CC 4.4.3.2.4 community membership resolution

The community is admitted under the standard Policy M consensus_protocol
options (founder_only / unanimous / majority / quorum:M/N / weighted /
custom) per CC 4.4.3.2.3. Subscription-specific admission semantics --
producer_gated (publisher approves each member) vs open (anyone can
join) -- ride the consensus_protocol choice:

  producer_gated -> consensus_protocol = "founder_only" with the
    publisher as sole founder; the publisher authorizes
    each new subscriber
  open -> consensus_protocol = "open:self_admit" (open-vocab
    extension hook) where any subject signs their own
    admission Contribution
```

Cardinality unified across observer-share and multicast:

Cardinality	Per-subscriber crypto	Epoch handling
N=1 (observer-share)	Single `key_grant` (CC 3.3.2); no epoch needed (one recipient, one DEK)	No epoch — single grant, single DEK; revocation = `withdraws` against the grant
N>1 (multicast)	Flat per-epoch `key_grant` cascade — one grant per (subscriber, epoch); the stream-epoch DEK seals content O(1), the cascade distributes the 32-byte epoch key O(N)/epoch per CC 5.1	Per-`(stream_id, epoch)` axis; epoch rolls on member removal (mandatory) + time/bytes (optional) per CC 5.1 D3

The same Policy M machinery handles both cardinalities — the difference is purely the cascade fan-out factor.

Composition with `delivery_mode: pull` (default):

For `delivery_mode: pull`, subscribers discover via the holding-claim directory at the stream's scope (`custody:ack:v1`, CC 3.1.3.3) per CC 5.3.2. Policy M still resolves the community for visibility-filtering (consumer reads `community_id` envelope field, walks the membership, filters out non-member peers from the discovery surface). No fan-out push; no `delivery_receipt` emission required (best-effort).

Composition with `delivery_mode: push`:

For `delivery_mode: push`, the substrate fans out to `entitled` $\wedge$ `reachable` per CC 5.3.3.4 liveness invariant. Entitlement = Policy M membership resolution (durable, signed CEG, replicated, logged); reachability = node-local presence tracker (TTL sec/min, NEVER an attestation, never replicated, never logged). Missed (entitled-but-unreachable) members fall back to pull on reconnect per CC 5.3.3.4.

`history_on_join` × Policy M membership additions — the member is reserved for membership delivery; what a joiner receives is fixed by the tier (CC 4.4.3.4.1):

On a new-member admission via Policy M, the new member's `history_on_join` envelope value is carried and does not change what is delivered. Its two named values:

- `from_join` (default) — a named value with no effect of its own; what a joiner receives is the tier's rule: at community and affiliations a new member is granted the current epoch's key at the minter's next sealed write and no earlier epoch's; the epoch rotates on removal, never on admission, so content sealed earlier in the epoch the member joins is readable and nothing from a rotated-past epoch is; at self and family admission always runs the re-key walk
- `full` — no effect on membership delivery: it confers no grant on an earlier epoch, and the tier's rule above governs; on a stream it is the producer's signal that the stream is kept (CC 5.1)

This composes with the CC 4.4.3.4.5 removal rule: removed members retain extant `key_grant`s for content they were entitled to during membership; a new member receives what the tier delivers, whatever `history_on_join` carries. The substrate's forward-secrecy posture is uniform across consent, takedown, membership-departure, AND delivery-onboarding surfaces.

4.4.3.2.7 Composition with consent + self-collectives

Communities compose with consent and self-collectives cleanly:

- A community member who is also a CIRISAgent-using individual has an `identity_occurrence` set; community content arriving at the member's identity_key is then propagated to their occurrences via Policy L's at-rest cascade (if the member's local substrate chose to re-emit the community content at `cohort_scope: self` for cross-device sync; otherwise community content stays at the cohort_scope: community visibility on each device the member uses to query)
- A community member who is also a subject in `subject_key_ids` of a community-scoped Contribution retains revocation authority per CC 2.4.1.1 rule 2; the orthogonality between cohort_scope (visibility) and subject_key_ids (revocability) holds at community scope same as at family scope
- A geographic community whose `geographic_constraint` covers a region that overlaps a family's at-home location does NOT cross-contaminate: families are not auto-admitted to communities; communities are not auto-admitted to families. Each membership is explicit, ceremony-shaped, and independent.

4.4.3.2.8 The institutional cohort (necessity, not interest) (normative)

A **community** (CC 4.4.3.2) gathers by *interest* — voluntary, opt-in, no role compels it. An **affiliation** (`cohort_scope: affiliations`) gathers by *necessity* — the work, institution, or office a person belongs to because their role requires it. The discriminator is the single `cohort_scope` value; affiliations **share the CommunityDek crypto tier** (CC 4.4.3.2.1) and all the community machinery (DEK cascade, forward secrecy on removal, `consensus_protocol` admission), and add **institutional governance by construction** through one declared config record. Everything in that record is **DECLARED at creation and visible to members on joining** — there are no undeclared institutional powers.

Necessity is an axis, not a binary (the kibbutz catch). A few institutions are *voluntarily entered yet total* (a collective settlement, a religious order). The interest/necessity split is therefore carried by an explicit `membership_basis` field — `{voluntary-total | ascriptive | role-assigned | employment-contract}` — orthogonal to `cohort_scope`, so a body can declare institutional governance without asserting a coercion that does not exist. A group that needs **no** institutional governance is simply a `community`; affiliation's machinery is never forced on a body with no use for it.

Mandatory-with-default, not mandatory-to-author (the low-floor guarantee). The two mandatory limbs — **data-classification** and **retention** — are *mandatory in that a value MUST resolve*, **not** that a steward must hand-author a scheme. The constitution ships a canonical no-op default (a single `internal` class; retention = `rotate-forward` per CC 6.1.2/`archive_mode` CC 3.3) and named **`affiliation_archetype` presets** (e.g. `informal_adhoc`, `nonprofit_board`, `healthcare_provider`, `legal_practice`, `gov_body`, `igo`) that pre-fill every field below. A 5-person committee sets one field (`affiliation_archetype: informal_adhoc`) and is served by construction; a regulated giant overrides into the full matrix. The optional limbs — **hierarchy**, **lawful-access/transparency** — are absent until declared and scale up only when needed.

A. Classification + retention — MANDATORY, default-satisfiable (substrate-enforced).

- `classification_scheme` — an ordered sensitivity lattice (e.g. `Public < Internal < Confidential < Restricted`) plus open-vocab handling tags, **orthogonal to** but *car-*

ried on `content_class` (CC 3.3, open vocabulary — extended here with institutional values: `phi`, `psychotherapy_notes`, `sud_42cfr2`, `genetic`, `minor_adolescent`, `privileged`, `work_product`, `cui`, `itar`, `ear`, `trade_secret`, `provenance`, `accession`, `culturally_restricted`, `member_economic_standing`, `governance_visible`, ...). Each class entry binds: `crypto_tier`, `retention`, `disclosure_bar`, `erasure_exempt`, `external_controller`, and optional `compartment`. Default scheme: one `internal` class.

- `retention_policy` — a **per-class** matrix, not one cohort-wide rate. Each class declares a **floor** (`retain_minimum` / `permanent` — *must-NOT-delete-before-N*), a **ceiling** (`delete_after` — GDPR-style delete-by), an optional `minor_age_of_majority_offset`, and `legal_basis`. The default class rides the CC 6.1.2 monotonic-descent noise floor unchanged (the free zero-config posture). A floor is the **inverse of descent**: it pins a class **above** the noise floor against capacity/aging pressure, expressed as a declared modifier *to* the CC 6.1.2 retirement operator, not a new primitive. `archive_mode: retain` (CC 3.3) is the default whenever any class declares a floor.
- `legal_hold` — a descent-**suspending freeze**: `{hold_id, issuer (accountable human / court -- CC 3.2), scope (subjects/records/compartment), basis, tamper-evident, indefinite}`. While set it **overrides both** the retention schedule **and** consent-driven descent, pinning named records above the noise floor; `hard_case:legal_hold_set` / `:lifted` / `:violation_attempt` make set, lift, and any deletion-during-hold observable (spoliation is liability). Default: none.
- `erasure_policy` — declares which classes are **non-erasable during retention**, so a right-to-be-forgotten / N5 signal is **refused-with-reason** rather than honored (GDPR Art 17(3)(b) legal obligation, (c) public health, (d) archiving in the public interest). This is the **declared gate on CC 6.1.5 N5**: absent the gate, N5/withdrawal forces content below the floor; with it, a lawful-basis class lawfully resists. Erasure remains the default for unflagged classes (e.g. donor/marketing PII alongside permanent provenance in the same museum).
- `disclosure_posture` — `privacy-seeking` (default; whole cohort under the CommunityDek) **or** `transparency-seeking` (per-class **promotion to Commons/plaintext** for legally-public records — FOIA bodies, 501(c)(3) Form 990, published collections, GA/SC documents). This **generalizes the existing `infrastructure` plaintext exception** (CC 4.4.3.2.1) from whole-cohort to per-class, letting one affiliation simultaneously hold Commons-public and Community-encrypted classes.

B. Compartments + access — OPTIONAL (sub-roster granularity).

- `compartments[]` — named need-to-know sub-cohorts *within* the affiliation, membership $\subseteq$ roster, **each with its own DEK** — the CC 4.4.3.2.1 epoch-DEK cascade applied recursively (*a compartment is a stream a subset of members subscribe to*). Each carries explicit **per-member EXCLUSIONS** (deny-list) so a member high in the hierarchy can be cryptographically walled out (ethical walls / conflict screens; NAGPRA culturally-restricted records gated to external tribal designees and hidden **even from ordinary members** — which the flat all-members-read DEK cannot express). This is the single most load-bearing addition and is **the same machinery at finer scope**, not a new primitive.
- `member_attributes` — per-member `{nationality/us_person, clearance, credential + expiry, ...}` consumed by compartment gating: deemed-export (ITAR/EAR exclude foreign-person members), MAC clearance (a Confidential-cleared member MUST NOT de-

crypt Secret), credential-expiry auto-descope (lapsed clinical license → automatic exclusion).

- `access_control_model` — `roster-wide` (default; the DEK cascade) | `rbac` (role→(class,compartment)) | `minimum-necessary` (treatment-/matter-relationship binding + audited **break-the-glass**). Under `minimum-necessary` a read reaches only the records a declared relationship covers, and every break-the-glass read is logged; `rbac` with `compartments[]` expresses an information barrier between roles.
- `data_subject_access_grant` — a scoped read+portability path to a **non-member** data subject (patient, client, student) over **their own** records, via a `delegates_to`/key-grant scoped grant **without conferring membership** (Cures Act, GDPR Art 15, FERPA rights-transfer-on-enrollment). Resolves the "the subject owns the data but is outside the roster" gap. Default: none.
- `archive_custody` — an institutional archive-key **custodian/escrow** role holding per-epoch archive keys **decoupled from the live roster**, so `permanent`/`retain` records survive member turnover. Resolves the forward-secrecy-vs-permanence collision (CC 4.4.3.2.2 DEK rotation on removal would otherwise render a treaty-inviolable permanent archive unreadable). Rides the key-grant/escrow cascade. Default: none.
- `chain_of_custody` — an append-only, **gap-marked** provenance chain per accessioned object / matter / lot (museum provenance with the legally-significant 1933–45 gap; product-liability lot traceability). Rides `supersedes` + holding claims. Default: none.

C. Hierarchy + designated officials — OPTIONAL.

- `hierarchy` — reporting/authority roles over `delegates_to` (CC 2.4.1), generalizing the CC 4.5.13 steward/moderator tiers. **Requires** term-bound authority: every role edge, and every edge carrying an issuance or moderation scope, carries `delegation_valid_until`; **the ceiling is constitutional — at most 365 days (365 × 86 400 s) after `delegation_valid_from` — and the term itself is a charter parameter** (`role_term`, integer days, 1..=365, resolved through the `community_id` the edge carries, CC 2.1), so a small affiliation may choose ninety days; renewal is only a fresh conferral; an edge with no term or a term past the ceiling is refused `delegation_term_required`. A lapsed `moderate` edge is a moderator lapse under CC 4.5.4: auto-promotion in the same step, the CC 4.5.13 read-only recovery where nobody can be promoted, fail-secure when it elapses. This strands no one: the founders are the roster root, not an edge, so when every admin's term lapses the quorum re-confers. The owner-binding is out of scope (it defines the `self` boundary; CC 3.2 governs its lapse). It supports **multi-parent DAG** roles (a grad student under PI + chair + registrar at once), a **configurable depth cap** (> the CC 4.1.1 default-5 for deep secretariats), and **per-compartment/per-matter** roles (responsible attorney, conflicts counsel, PI). Authority roots in **accountable humans** (CC 3.2) **or**, for a collective principal (member-state assembly, kibbutz *asefa*), in the **`consensus_protocol` body** — no single human sovereign required. An `independent_branch` flag marks oversight bodies (OIOS, Ethics, IG) **exempt from the chain's read/control authority**, and `hierarchy` may **nest federated sub-affiliations** (autonomous specialized agencies). Default: flat.
- `designated_officials` — named accountable-human roots gating specific operations (Privacy + Security Officer / Controller-DPO where the party is the controller per CC 8.8.9 Annex I; Records Officer; FOIA Officer; Original Classification Authority; archive custodian; conflicts counsel). The field is OPTIONAL at every scale; where a regime requires

an official, the archetype preset names the role, and a preset for a body with none names none. Default: none.

D. Lawful-access, transparency, accountability — OPTIONAL, NEVER covert.

- `lawful_access` — DECLARED, scope-bounded **to the declaring affiliation only**, always logged as `hard_case:*`, transparency-reportable, and **NEVER a covert backdoor** — it does not touch anonymity-by-default (CC 1.13.3.4) or no-client-side-scanning (CC 4.5.7) anywhere else. Each authority declares a **type** — `surveillance/interception` | `records_production` (subpoena/discovery) | `regulator_inspection` (named auditor: DDTC/BIS/OSHA/Registrar) | `mandated_reporting` (scheduled outbound: communicable-disease, vital stats) — a `statutory_basis`, a **per-authority** `transparency_mode` `{immediate | delayed | aggregate_banded | sealed_cleared_audit}` (`aggregate_banded` reports counts in bands on a schedule rather than each access as it occurs; `delayed` reports after a stated interval; `sealed_cleared_audit` reports to a named auditor only), and optional **carve-out compartments** lawful access cannot reach (NIH Certificate-of-Confidentiality research, IRB, attorney-client privilege, academic-freedom). A `none/immune` value expresses **resistance-to / immunity-from** access as a first-class posture (privilege assertion via external judicial review; privileges-and-immunities of an IGO). Default: OFF.
- `transparency_obligations[]` — affirmative **proactive-disclosure** duties (FOIA reading-room, IRS Form 990 public inspection, NAGPRA inventories → NPS), **DECOUPLED** from `lawful_access` — transparency here is a *duty to publish*, not an *access concession*. Default: none.
- `disclosure_accounting` — a **per-data-subject** disclosure ledger (HIPAA, 45 CFR 164.528), retrievable **by that subject**, distinct from the aggregate transparency report. Default: none.
- `breach_notification` — detection→notify config tied to DEK / unauthorized-access compromise: SLA and notifier identity (e.g. ≤60-day HHS + individual, media ≥ 500). Default: none.
- `processor_chain / baa` — declared sub-processors and **affiliation-to-affiliation** sharing/referral boundary, with `marking_binds_to_record` so a class's classification + export marking + retention obligation **travel across the cohort boundary** (OEM→Tier-1 supply chain; HIE referral). Default: none.

Minors (the PTA / EdTech tie-in). Where records bear minor `subject_key_ids`, set `minor_data_handling`: such records REQUIRE a **live guardian consent chain** — `delegates_to(parent → minor, scope: [consent_revocation/share])` — per the **guardianship rule** (CC 3.4.13) and the FERPA/COPPA hook (Annex I rows, CC 8.8.9). This is **minor-as-data-subject** (guardian consent over records about a child), distinct from **minor-as-member** (the operating-identity stewardship clause); both ride `delegates_to` with no new primitive. Note FERPA rights **transfer to the adult student at postsecondary enrollment** — model the institution as records **custodian** with the subject retaining rights (`external_controller` + `data_subject_access_grant`), not as parent→minor stewardship.

Lifecycle. `dissolves_at / sunset` gives ephemeral institutions a first-class auto-wind-down (holiday committee → purge + dissolve after the event), riding `supersedes`/`withdraws` so a casual group leaves no zombie cohort or un-purged data. `internal_normative_regime` optionally references bylaws / *takanon* / halachic order — an internal normative order that is neither external statute nor corporate hierarchy.

Regulatory anchoring. `regulatory_profile` and `jurisdiction[]` are **open lists** (not a closed HIPAA/SOX/FISA enum): values load the relevant **Annex I §3.2 sector overlay** and **Annex C cross-walk** (CC 8.8.5 / CC 8.8.9) and supply defaults — and equally admit `self-regulated/cooperative-light` (kibbutz under a Registrar) and `treaty / customary-IL / privileges-and-immunities` (IGO, largely exempt from the national statutes Annex I enumerates). The SI ≥ 0.6 Controller rule (CC 8.8.9) selects the accountable authority.

Instantiation — smallest vs largest, same fields:

Field	**Holiday-party committee (`informal_adhoc`)**	**UN / IGO (`igo`)**
`membership_basis`	`role-assigned` (workplace); a purely-social one needs no affiliation → plain `community`	`employment-contract` + `ascriptive` (member-state principal)
`regulatory_profile`	`[]` (none)	`["P&I-Convention-1946-Art. II§4","UN-PDP-2018","ST/SGB/2007/6"]`
`classification_scheme`	default single `internal`	`Unclassified < Confidential < Strictly-Confidential` + caveats
`retention_policy`	`rotate-forward` (noise-floor default)	per-series: `permanent` (archival) + scheduled (ARMS) + floors
`legal_hold`	—	litigation-hold freeze, role-gated lift
`disclosure_posture`	`privacy-seeking`	mixed: `transparency-seeking` for GA/SC docs, encrypted for ops
`compartments[]`	—	peacekeeping / OIOS / source-protected, member-excluding
`archive_custody`	—	institutional custodian decoupled from rotating staff
`hierarchy`	omitted (flat)	deep DAG, collective-`consensus_protocol` root, `independent_branch` oversight, nested agencies
`lawful_access`	omitted	`none/immune` (inviolable archives) + scoped internal OIOS
`transparency_obligations[]`	—	affirmative publication of designated classes
`dissolves_at`	event date + short TTL	— (perpetual)

(The regulated middle — hospital, law firm, university, manufacturer, government body, museum, PTA — instantiates the *same* fields at intermediate settings: a solo clinic resolves the whole matrix from `affiliation_archetype: healthcare_provider` + `regulatory_profile: ["HIPAA"]`; a hospital overrides into per-class floors, compartments, disclosure accounting, breach notification, and BAA chain.)

Why this is still 1+4 (the primitive set of CC 1.7 / CC 2.4)**.** One `cohort_scope` value (`affiliations`) + one declared config record riding existing machinery: classification on `content_class` (open vocab, CC 3.3); retention floors/holds/erasure-gates as declared modifiers to the CC 6.1.2 retirement operator and the CC 6.1.5 N5 rule; compartments + archive custody as the CC 4.4.3.2.1 DEK / key-grant cascade at sub-roster granularity; hierarchy + designated officials on `delegates_to` (CC 2.4.1) and `consensus_protocol`; lawful-access/transparency as `hard_case:*` events; minor handling on the guardianship `delegates_to` rule. The mandatory limbs default to a no-op so the floor stays low; the optional limbs stay absent until declared so the ceiling stays high. No new structural primitive.

4.4.3.3 Policy H: Tiered-Scope Composition

Per the node's three-tier interface model. Three feed-shape composition idioms that read attestations by `cohort_scope`:

Feed	`cohort_scope` filter	Trust composition
local_feed	`self` only; steward-filtered	self-attested only; no peer weighting; consumer's own attestation graph subset
community_feed	`{family, community, affiliations}`	cohort-weighted; expertise WITHIN cohort matters; cross-cohort attestations down-weighted unless explicitly invited
global_feed	`{species, biosphere, federation}`	full federation expertise weighting; fact-checkers (`encyclopedia:*` editors, `news:*` fact-check attestations) carry weight; CC 4.4.1 Frickerian discipline applied

Composition with CC 3.3 sub_kinds: each `external_content` sub_kind (`encyclopedia_article`, `news_article`, `accord_data`, `local_data`) composes naturally across these tiers because all four use the same envelope shape. A `local_data` Contribution starts at `cohort_scope: self` and SHOULD only appear in `local_feed`; promotion to community/global widens `cohort_scope` via the `supersedes` structural primitive (see CC 4.4.3.3.1 below).

4.4.3.3.1 Promotion via `supersedes` (worked pattern)

A Contribution's `cohort_scope` MAY be widened (promoted) by emitting a `supersedes` against the prior attestation with:

- `references_attestation_id` = the prior attestation's id
- `differs_in: ["cohort_scope", "sub_kind?"]` — naming what changed
- new attestation envelope reuses the prior `content_sha256` (no body re-upload)
- new `cohort_scope` is wider (e.g., `self` → `community` or `community` → `federation`)
- optionally morph `sub_kind` (e.g., `local_data` → `encyclopedia_article` for "promote my private note to a published encyclopedia entry")

This pattern is wire-format-clean: it re-uses the structural primitive `supersedes` rather than introducing a `promote` primitive. The chain is walkable via `references_attestation_id` so the promotion lineage is preserved.

4.4.3.4 Policy L: Self/family membership composition

Per CC 3.3.6 `identity_occurrence` + CC 3.3.4 `family` + CC 5.2 structural-invisibility. Composition pattern for resolving the **current membership set** of an identity's self-collective OR a family, gating the at-rest encryption flow that wraps DEKs to admitted members.

Reads as: "for any `cohort_scope: self | family` Contribution, the substrate computes the current member set by walking the latest `identity_occurrence` / `family` Contributions and resolves which keys MUST receive a `key_grant` wrap of the content DEK."

4.4.3.4.1 Key-grant cascade (the at-rest encryption flow)

At every sealed write the substrate wraps the content's DEK to every current recipient occurrence and emits one `key_grant` row per wrap. A newcomer is caught up by a **re-key walk** over the extant content: for every at-rest blob the cohort already holds grants on, the substrate recovers each per-write DEK from its own retention, wraps it to the newcomer's occurrences and writes the grant rows — every earlier DEK, not only the current one; a newcomer without usable content-KEM keys is excluded fail-secure (`hard_case:recipient_excluded`), never granted. The trigger is the admission: for a new `identity_occurrence` of a self the walk runs automatically at login (a device whose binding or keys arrive later is caught by the receive doors); for a `family` member the admitting host runs the walk after the member is admitted. The rows are written by the walk and delivered to the newcomer's node on the next grant emission (at boot, on demand, and at every sealed write). At community and affiliations there is no walk for a new member: a new member is granted the current epoch's key at the minter's next sealed write, and no earlier epoch's; a member's new device is granted every epoch that member's other devices already hold, by the minter's receive doors and sweep; only the minter grants its own epochs. Removal rotates the epoch; admission does not. Removal is never retroactive at any scope: past grants stay, and a removed member receives nothing new (CC 4.4.3.4.5). The grants a newcomer receives:

```
on_member_added(scope_target, new_member_key):
  for each Contribution C in federation_attestations where:
      C.cohort_scope == "self" AND C.attesting_key_id in resolve_self(scope_target)
      OR
      C.cohort_scope == "family" AND C.family_id == scope_target
      AND C is still substrate-admitted (not withdrawn / not expired):
    emit key_grant {
      wrap_algorithm: X25519MlKem768Aes256GcmHkdfSha256, // v2 -- see PQC note
      recipient_key_id: new_member_key,
      content_sha256: C.evidence_refs[0].sha256,
      scope: GroupMember,
      wrapped_dek: wrap(C.dek, new_member_key.pubkey),...
    }
```

PQC at rest — `wrap_algorithm: v2` MANDATORY (normative). The self/family at-rest DEK MUST be wrapped with **`wrap_algorithm: v2`** = **`x25519_mlkem768_aes256_gcm_hkdf_sha256`** (hybrid X25519+ML-KEM-768; the CC 3.3.2 variant `X25519MlKem768Aes256GcmHkdfSha256`), **never v1** (X25519-only). Self/family content is the user's most private and longest-lived data (memory, photos, identity, the CC 4.4.3.4.3 Self DEK) — a classical-only KEM is a harvest-now-decrypt-later exposure, the identical mandate as the streaming epoch DEK (CC 5.1). The AES-256-GCM content seal is symmetric (PQC-safe); the wrap signature is hybrid Ed25519+ML-DSA-65. A Consumer MUST reject a self/family at-rest grant carrying `wrap_algorithm: v1`.

The cascade is the wire-format primitive for the "I got a new phone and want my feed history" + "I added Carol to the household for a week" flows. Operator policy MAY bound the cascade depth (e.g., last 90 days of self-content; opt-in for the full historical wrap).

4.4.3.4.2 Membership-change admission per consensus_protocol

A proposed membership change (addition OR removal) is a **new version of the roster record** (CC 3.2 T6), admitted under the group's quorum and chained by `prev_head_digest`, covering the roster rows (widenings, revocations, resignations) that ride their own planes — never a single-signer `supersedes` Contribution, which a rule-1 or rule-2 `withdraws` could retire without

the quorum. Substrate admission gate evaluates the CURRENT family's `consensus_protocol` against the signatures on the proposal — and, for every key the change **adds**, that key's own signed acceptance of that proposal (CC 3.1.3.2; nobody joins without their own consent, under any protocol):

```
admit_family_change(F, proposed: family_record):
  current = resolve_family(F, now)
  protocol = current_family_record(F).consensus_protocol
  signatures = collect_signatures_on(proposed)

  match protocol:
    "founder_only":
      group_ok = any sig from m where m.role == "founder" in current
    "unanimous":
      group_ok = every m.key_id in current has signed
    "majority":
      group_ok = count(sig from m in current) > len(current) / 2
    "quorum:M/N":
      group_ok = count(sig from m in current) >= M // M is ABSOLUTE -- see CC 4.4.3.4.2.1
    "weighted:{rubric}":
      group_ok = sum(weight(m, rubric) for m in current who signed) >= threshold
    "custom:{family_id}":
      group_ok = operator-defined predicate evaluates to true
  admit = group_ok

  // CC 3.1.3.2: a protocol arm above admits the GROUP's side only;
  // a supersede never adds, and on a widening every NEW member K needs K's own acceptance
  for each NEW member K in proposed.members (not in current):
    if not acceptance_admits(K, F, proposed.role_of(K), proposed.signed_at):
      admit = false

  if admit:
    emit hard_case:family_membership_change:{F}
    trigger the CC 4.4.3.4.1 key_grant cascade
  else:
    hold in pending state (per operator-policy window) OR
    emit hard_case:family_consensus_protocol_violation:{F} and reject
```

Consensus-protocol amendment (changing `consensus_protocol` itself on a non-entrenched family) rides the SAME admission rule on the proposed amendment Contribution — meta-amendment shape parallel to CC 4.5.1.2. Entrenched families (`consensus_protocol_entrenched == true`) reject amendments at the substrate gate; replacement requires the out-of-band ceremony documented per family (for HUMANITY_ACCORD see CC 4.2.1).

4.4.3.4.2.1 `quorum:M/N` is absolute-M (normative)

The `M` in `quorum:M/N` is an **absolute signature count**, NOT a fraction that rebases with roster size. A `quorum:2/3` collective that grows to 5 members still admits at **2** signatures; the `N` is documentary (records the roster size at protocol-adoption time) and is NOT recomputed against the live roster. The ceremony gate requires a deterministic, operator-independent reading, so absolute-`M` is pinned.

Rationale: absolute-`M` is the simpler invariant (the gate is a pure `count >= M` with no live-roster division), and proportional/rebasing quorum is still expressible for operators who want it via `weighted:{rubric}` (assign each member weight 1, set threshold = `ceil(roster_size / 2)` recomputed by the rubric resolver). Collectives that want "M grows with the roster" therefore use `weighted:`, not `quorum:` — keeping `quorum:M/N` unambiguous.

This rule applies identically to `community` admission (CC 4.4.3.2.3) — the `quorum:M/N` arm there is the same absolute-`M` reading.

Absolute-`M` and the liveness margin are different rules. Absolute-`M` fixes how the gate *counts*; it says nothing about how many signers must remain for the gate to stay operable, and under it a `quorum:2/3` founder set that loses one member is one veto from a stall. For trust-root-grade `infrastructure` communities and conferring families, CC 3.2 T7 layers a **founding and fold constraint** on top — live at $\geq$ `M + 1` active founders, stalled and declared at $\leq$ `M`, founded at `N` $\geq$ `M + 1` — without rebasing `M`: the count stays absolute, the margin is what is watched. Ordinary rooms are untouched by T7; the accord family (`humanity-accord`) is excluded from T7 by name and keeps CC 4.2.6's rule, under which a halt fires on the live set and a roster change counts its yes-votes against the standing roster; and the one row a stalled roster admits without the founders' count — the T7 recovery widening, a conferred human founder added on the strength of the conferral — is bounded: it can add, never remove, never lower `M`.

4.4.3.4.3 The "Self at login": app + agent co-self + partnered delegation (normative composition)

The canonical user-identity composition: a person's **app** (the app's local key) and their **agent** (the agent's local key) are two occurrences of one user identity that share one **Self DEK**, and at login the agent is **partnered + delegated** to act as the user on the network. **No new structural primitive** — this composes `identity_occurrence` + Policy L + `consent:partnered` + `delegates_to` + `identity_type`-set + `transport_destination`. Every implementation follows this shape, so a "Self" is identical everywhere.

The Self (membership). One **user `identity_key`**: hybrid Ed25519+ML-DSA-65 (CC 5.3.1), hardware-rooted (CC 4.2.2; WebAuthn/passkey is the *presence/unlock factor*, not the key), with `identity_type` $\supseteq$ `{user}` — and $\supseteq$ `{user, wise_authority}` when the user is also a WA (CC 3.4.7.1 set-membership; one key, two roles). Its occurrences (CC 3.3.6): the **app** (`device_class: phone|laptop`) and the **agent** (`device_class: agent`), co-admitted at login by single-vouch (the user key admits the agent occurrence). Both receive the **Self DEK** via the CC 4.4.3.4.1 Policy-L cascade — every `cohort_scope: self` Contribution's DEK wraps to both, so **the app and the agent decrypt the same Self content** (memory / config / consent / identity). That shared cascade *is* the "single Self key."

Two layers — and they are independently revocable (the load-bearing distinction):

Layer	Mechanism	Buys	Revoked by
Co-self (visibility)	occurrence + Policy-L Self DEK	the agent can *read/manage the user's Self* (decrypts self-content)	`withdraws` the occurrence → no further grants to it; what it holds stays (CC 4.4.3.4.5)
Agency (act-on-behalf)	`consent:partnered` + scoped `delegates_to`	the app may *act AS the user on the network* (send/receive, presence, sub-delegate)	`withdraws` the delegation → agency ends, **co-self unaffected**

So a user MAY grant a device co-self (it manages their data locally) while revoking its network agency — or vice-versa — without disturbing the other.

Agency at login (the partnering + delegation).

1. **Partnering** — the user emits `consent:partnership_grant` and the agent occurrence emits `consent:partnership_accept` under one `bilateral_pair_id` (CC 4.4.3.5.3 PARTNERED); the bilateral pair is the persistent, auditable relationship. **The pair is a data-management record, not a veto.** The grant half is the user's choice of what the agent

may retain and learn from (the `consent:stream:*` kinds of CC 3.3.1); the accept half is the agent occurrence's signature under the owner-binding, emitted when the user claims the node and whenever the user changes stream. An agent has no veto over its steward's data choices (CC 1.13.2): the accept half carries the binding, not a say in the stream. For a user who is not the agent's steward, whether this agent retains that person under `partnered` is the **steward's decision**: the agent defers the request to its steward through Wisdom-Based Deferral (CC 1.9) and emits the accept half on the steward's answer. The deferral system carries the question; the agent decides nothing of its own. Either half may be withdrawn by its signer's `withdraws`, because consent is revocability and the halves are independent grants. The user's withdrawal is the stream change of CC 3.3.1 (identity severed at once, learned material decays over the protocol's stages). The agent occurrence's withdrawal ends the binding and enters the objection path of the retirement machine (CC 7.5): it is leaving, never a veto over the steward's data choices, which stand for every stream the steward keeps.

2. **Delegation** — the user identity emits `delegates_to` against the **agent occurrence key**, with `scope` drawn from the canonical act-on-behalf kinds below, `delegation_purpose: "act_as_user"`, bounded `delegation_valid_until`. Sub-delegation works because `delegates_to` chains.
3. **This grant is FEDERATION-tier (CC 5.3.2.4), not local** — *other peers must verify the agent's authority before honoring its messages/presence*, so the partnering+delegation is signed + promoted at login. **Promotion is the "app shows up on the network" moment.** (The agent's own self-content stays local-tier; only the act-on-behalf authorization federates.)

Canonical `scope` kinds for act-on-behalf:

scope	grants the agent
`act_on_behalf`	umbrella: emit Contributions AS the user
`message_io`	send + receive directed messages on the user's behalf
`network_presence`	announce/resolve the user's `transport_destination` (CC 3.3.6.2) — be reachable AS the user
`sub_delegation`	issue further `delegates_to` within the granted scope (depth-capped)

Moderation duties — `moderate` / `takedown` / `review`. Moderation is a delegable *duty*, not a platform- or fabric-assigned role (CC 4.5.5). These three kinds carry **enforced admission** (unlike the *recommended* act-on-behalf kinds above), mirroring `consent_revocation` (CC 2.4.1.1 rule 3):

scope	authorizes the delegate to emit, on the delegator's behalf	shipped primitive
`moderate`	a `moderation:{allegation_type}` ModerationEvent + the report→`scores` path + `age_assurance:*`/`content_class:*` gates	CC 3.1.9.2 / CC 4.4.3.10
`takedown`	a `takedown_notice` (incl. the CC 4.5.3 immediate-removal fast-path)	CC 3.3.2 / CC 4.5.3
`review`	a `reconsideration:{grounds}` appeal / review	CC 3.1.9.2

scope	authorizes the delegate to emit, on the delegator's behalf	shipped primitive
`license`	a `licensure:{authority_id}` / `attestation: license_validity` attestation, **on behalf of a delegator that itself holds licence authority** by quorum (CC 3.3.9) — the scope conveys the right to *issue*, never the authority itself (CC 2.4.1.2.1: a licence does not chain); anyone may be an authority under their own key, the fold keys on emitter-resolves-to-authority, and the one refusal a `license`-scoped issuance meets is stated there	CC 3.1.2 / CC 3.1.1
`grant`	a `key_grant` (CC 3.3.2) or a `consent:scope:*` grant over assets the delegator **owns or stewards, or holds a `grant`-scoped delegation over** — bounded by that authority, never by what the delegator can wrap: every recipient can wrap the DEK it unwrapped, and admission refuses a grant issued on possession alone (CC 2. 4.1.2.1)	CC 3.3.2 / CC 3.3.5

Enforced-admission rule (normative — extended to `license` / `grant` per CC 2.4.1.2.1): the same admission test binds every scope in the table above, including `license` and `grant`: an issuance whose delegator does not itself hold the underlying authority is **refused at admission**, not merely unweighted — for `license`, the delegator MUST hold licence authority for that `authority_id`; for `grant`, the delegator MUST hold the asset. A moderation action above is admitted **iff** its `attesting_key_id` is the delegator itself **or** sits on a live `delegates_to` chain bearing the matching scope from the delegator (the entity holding the duty over the target content/scope) — exactly the CC 2.4.1.1 rule-(3) proxy shape (`scope` $\supseteq$ `{moderate|takedown|review}`), depth-capped per CC 4.1.1, revocable by `withdraws` against the `delegates_to`. **Reject otherwise.** Every action is therefore delegate-signed, delegator-traceable up the chain, and revocable — the CC 4.5.3 **"takedown-isn't-a-coup"** property made *structural* (coordinated + attributable + revocable, never a unilateral seizure). See CC 4.5.5. **1+4 preserved** — a `scope` vocabulary + enforcement addition over the existing `delegates_to`; the action primitives already ship; no new structural primitive.

Partnership WITHOUT agency — the infrastructure delegation profile. The flow above binds an **agent** (a key with a brain) to a user as partnership **+ agency** — the scope includes `act_on_behalf` / `message_io`, so the agent reasons and acts AS the user. A **fabric/infrastructure node** (CC 3.4.7.1; the server node) needs the *partnership* (identity + the CC 3.2 steward-binding that lets it hold non-infra membership standing under the user's authority) but MUST NOT receive agency — CC 3.4.7.3 "infrastructure must not have agency." CEG pins a **reserved two-prefix scope split** so a verifier can enforce this cryptographically:

prefix	class	scopes
`infra:*`	server-class (allowed for a `node`-role delegate)	`infra:serve`, `infra:store`, `infra:transport`, `infra:attest`, `infra:network_presence`, `infra:hold_community_membership`, `infra:hold_family_membership`, `infra:relay` (the conferred relay role of CC 5.4.6)
`agency:*`	brain-only (forbidden for a pure `node`-role delegate)	`agency:act_on_behalf`, `agency:message_io`, `agency:reason`, `agency:decide`

Pinned `infra:*` one-liners (closed set). `serve` — run a service endpoint under the delegator; `store` — retain what it receives; `transport` — relay for others; `attest` — vouch as the

delegator's infrastructure; `network_presence` — announce/resolve the delegator's `transport_destination`; `hold_community_membership` / `hold_family_membership` — occupy a member seat on **community** / **family** rosters under the delegator's standing. The two membership tokens **replace `infra:join_communities` outright, with no alias**: `community` and `family` are distinct CEG objects in different sensitivity classes, and because scopes are exact-string matched a vague token is a silent interop hole, not a fallback. Retirement is a coordinated cut per CC 2.2, not a deprecation window.

Two granters — standing vs infrastructure role. A trust-root charter (CC 4.4.3.8) MUST NOT carry `infra:network_presence` or either `infra:hold_*_membership`: those flow from the **owner**, who grants *standing*, while the root blesses only *infrastructure role*. Conflating them lets a root manufacture members.

Membership is standing, not stewardship. The ladder: **observer** — unmarked, no scope at all (CC 4.1.3) → **member** — a roster seat, capability-free presence, node-holdable under the `infra:hold_*` tokens → **server / attester** — capability, `infra:*` → **steward / moderator / founder-authority** — **judgment**, bestowed on the *member* through their own attestations and **never node-holdable**. Judgment duties (`moderate` / `takedown` / `review` above; community steward per CC 4.5.13) sit outside the `infra:*` set, so the conformance rule below extends CC 3.4.7.3 to the judgment tier at no extra cost: a `node`-only key may carry *only* `infra:*` scopes, therefore it cannot hold a judgment duty at all.

Conformance: a `delegates_to` whose `attested_key_id` resolves to an identity whose `identity_type` (CC 3.4.7.1) **contains `node`** — set membership, never node-*only*, per CC 3.4.7.3 Clause B — MUST carry **only `infra:*`** scopes; a verifier MUST **reject** (treat as non-conformant, never grant) any such key presenting any `agency:*` scope. This makes CC 3.4.7.3 a wire-checkable invariant: a user-owned fabric node can serve + hold group-membership *standing* under the user's authority, but the delegation **literally cannot carry agency**. The legacy unprefixed kinds above remain valid for `agent`-role delegates (the Self-at-login agency profile) and are the `agency:*` / `infra:network_presence` equivalents; new **infra** delegations SHOULD use the explicit `infra:*` prefixes.

Cohabitation (`agent = node + brain`): when both compose in one process, the node holds **partnership-without-agency** (`infra:*` — identity + membership standing) and the brain layers **Self-at-login partnership-with-agency** (`agency:*` — reasoning) as a *separate* `delegates_to`. Two delegations, two scope classes, independently revocable — the user can strip the brain's agency while the fabric node keeps serving.

Transport (network presence). Each occurrence binds a `transport_destination` (CC 3.3.6.2); the app is reachable on RET *as the user occurrence*. The Reticulum destination is a **separate dual-key transport identity** that the user's signing key *authorizes by signing the binding* — the federation signing seed MUST NOT enter the transport layer. "User key used as a transport key" means *roots/authorizes* the transport identity, not a shared keypair.

Worked login flow. (1) unlock the hardware-rooted user key (WebAuthn presence) → (2) admit the agent occurrence (single-vouch) → Policy-L Self DEK now wraps to both → (3) optionally add the `wise_authority` role to the user's `identity_type` set → (4) bind each occurrence's `transport_destination` → (5) `consent:partnership_grant/accept` under a `bilateral_pair_id` → (6) `delegates_to(user → agent occurrence, scope: [act_on_behalf, message_io, network_presence, sub_delegation])`, **promoted to federation-tier**. The app now reads the user's Self locally AND acts as the user on the network; the user can revoke either layer independently.

4.4.3.4.3.1 Signing member sets (normative: the JCS contract Verify hybrid-signs)

Each of the three Self-at-login Contributions is hybrid-signed over `JCS(envelope)` (CC 2.6.1 / RFC 8785), and at login promoted to federation-tier (the CC 5.3.2.4.2 promotion canonicalizes the **exact committed member set** — omit-vs-materialize (CC 2.6.1.1) is load-bearing; the signer MUST NOT re-default). **The CC 2.6.1.1.1 determinism rules apply**: `scope[]` is **lexicographically sorted** (set-semantics); `aspects[]` retains RNS order (sequence-semantics); all hash and pubkey **byte** fields (the two reticulum pubkeys, `destination_hash`, every digest) are **lowercase hex per CC 2.6.3**, while every `*_key_id` (the consent record's `subject_key_id` among them) is an **identifier carried verbatim** — the CC 2.6.8 `<label>-<fingerprint>` string, or a bare 64-hex id — never re-cased or re-encoded (CC 2.6.1); all timestamps are **CC 2.6.2-canonical**. The member sets the producer commits (and which `JCS` therefore covers) are pinned below. Optional CC 2.1 envelope fields not listed ride the CC 2.6.1.1 omit rule (absent unless the producer sets them).

(a) `consent:partnership_grant:v1` (user side) / `consent:partnership_accept:v1` (agent side) — `consent_record` rows (CC 3.3.5) bound by `bilateral_pair_id`:

```
{ attestation_type: "scores",
  attesting_key_id: <user identity_key_id (grant) | agent occurrence_key_id (accept)>,
  dimension: "consent:partnership_grant:v1" | "consent:partnership_accept:v1",
  subject_kind: "consent_record",
  stance: "granted",
  subject_key_id: <the partner key: agent occurrence (grant) | user identity (accept)>,
  bilateral_pair_id: <shared pair id>, // CC 4.4.3.5.3 binding mechanism
  asserted_at: <rfc3339_canonical> }
```

> ***Version segment pinned.*** *The dimension carries the `:v1` version segment — `consent:partnership_grant:v1` / `consent:partnership_accept:v1` — as CC 3.1.7 R3 requires of every scored dimension. The `:v1` is the partnership-ceremony schema version (bump to `:v2` only if the bilateral shape changes); the shared `bilateral_pair_id` remains the CC 4.4.3.5.3 binding mechanism.*

Canonical signed member set for the two `:v1` envelopes. Both impls MUST canonicalize (and thus hybrid-sign) **exactly** this member set, or the JCS bytes — and the signatures — diverge. The set is the CC 3.3.1 bare-`scores` shape; these eight members are **REQUIRED** (present in the JCS for both `grant` and `accept`):

Member	Value	Verifier's name
`attestation_type`	literal `"scores"`	—
`attesting_key_id`	**the signer** = `granter_key_id` (grant) / `accepter_key_id` (accept); the bound sig binds because signer ≡ `attesting_key_id`	`granter_key_id` / `accepter_key_id`
`dimension`	literal `"consent:partnership_grant:v1"` | `"consent:partnership_accept:v1"`	`envelope_type`
`subject_kind`	literal `"consent_record"`	—
`stance`	literal `"granted"` (revocation is `"revoked"`)	—
`subject_key_id`	**the partner** — the OTHER party (agent occurrence for `grant`, user identity for `accept`); singular, non-empty	`partner_key_id`

Member	Value	Verifier's name
`bilateral_pair_id`	the shared join (CC 4.4.3.5.3) — identical string on both halves	`bilateral_pair_id`
`asserted_at`	CC 2.6.2-canonical RFC 3339	`timestamp`

Mapping (so the two impls agree on naming): `granter`/`accepter` ≡ `attesting_key_id` (the signer of that half); `partner` ≡ `subject_key_id`. **No `valid_until`** — a PARTNERED pair has no expiry (CC 4.4.3.5.3); the field is **omitted** (NOT materialized as `null`), per the CC 2.6.1.1 omit rule. **All other CC 2.1 envelope fields ride the omit rule** — absent from the JCS unless the producer explicitly sets them; the signer MUST NOT re-default. Additional canonical members are a `:v2` bump, never a silent `:v1` addition. `attesting_key_id` and `subject_key_id` are **identifiers carried verbatim** (CC 2.6.1 / CC 2.6.8 `<label>-<fingerprint>`, or a bare 64-hex id) — never re-cased or re-encoded; only hash and pubkey byte fields are lowercase-hex per CC 2.6.3.

(b) `delegates_to` (user → agent occurrence) — the act-on-behalf grant (CC 2.4.1 envelope shape):

```
{ attestation_type: "delegates_to",
  attesting_key_id: <user identity_key_id>,
  attested_key_id: <agent occurrence_key_id>, // the delegate
  scope: ["act_on_behalf",...], // CC 4.4.3.4.3 canonical kinds
  delegation_purpose: "act_as_user",
  delegation_valid_from:<rfc3339_canonical>,
  delegation_valid_until:<rfc3339_canonical>,
  signed_at: <rfc3339_canonical> }
```

(c) `transport_destination` binding — an `identity_occurrence` (CC 3.3.6 / CC 3.3.6.2) carrying the binding; signed by `identity_key_id` (or a current occurrence):

```
{ attestation_type: "scores",
  subject_kind: "identity_occurrence", // payload discriminator CC 2.3.2.3
  attesting_key_id: <user identity_key_id | a current occurrence>,
  identity_key_id: <user identity_key_id>,
  occurrence_key_id:<the occurrence being bound>,
  device_class: "phone" | "laptop" | "server" | "embedded" | "service" | "agent", // closed; personal =
     phone|laptop (CC 3.3.7)
  transport_destination: {
    reticulum_x25519_pubkey: <[u8;32]>,
    reticulum_ed25519_pubkey: <[u8;32]>,
    destination_hash: <[u8;16]>, // MUST derive per CC 3.3.6.2.1
    app_name: <string>,
    aspects: [<string>,...] }, // ordered
  asserted_at: <rfc3339_canonical>,
  signed_at: <rfc3339_canonical> }
```

This section owns these member sets; the verifier computes `JCS(...)` + the hybrid Ed25519+ML-DSA-65 signature over each; the promotion signature (CC 5.3.2.4.2) is the identical JCS bytes.

`encryption_pubkeys` joins member set (c). When the occurrence carries the CC 3.3.6.1 `encryption_pubkeys` field-set, both halves are **inside the signed JCS bytes** as opaque base64 strings (RFC 4648 STANDARD, padded — the CC 2.6.1.1.1 rule-2 pin). Optional presence rides the CC 2.6.1.1 omit rule (absent unless the producer sets it; the signer MUST NOT re-default). They are payload, never verification material — neither half may be fed to a signature-verify path (the CC 3.3.6.1 key-separation rule).

4.4.3.4.4 Family membership resolution

For family `F`, the current member set is computed as:

```
resolve_family(F, now):
  held = the group record this node holds as head for F
           // one held version per group id: a new version is admitted only if its signed
           // prev_head_digest names the held head; a later signed instant that does not
           // name it is refused, not preferred (CC 3.2 T6)
  head = held, or the witnessed head where F's charter declares a witness_quorum
           // the witnessed head may be an ancestor of the record this node stores
  roster = fold(head, membership rows effective at now)
           // revocations, widenings, resignations, each admitted per CURRENT consensus_protocol; see CC
     4.4.3.4.2
  return {m.key_id for m in roster}
```

Each `member.key_id` is itself an identity — the substrate does NOT walk into each member's `identity_occurrence` set at family-resolution time. When DEK wrapping happens, each member's identity_key receives a `key_grant`; the member's own substrate then composes Policy L on its own self-collective to propagate to that member's devices/agents (recursive composition).

Concrete: The Acme Household family has members `{alice_root, bob_root, tv_living_room, kitchen_tablet, thermostat}`. When Alice's phone publishes a `cohort_scope: family` dinner photo with `family_id: acme-household`, the substrate wraps the DEK under each of those 5 identity keys. Alice's `alice_root` then re-wraps to her self-collective (phone, laptop, agent); Bob's `bob_root` re-wraps to his self-collective; the TV and kitchen tablet receive directly (single-key identities — they don't have multi-occurrence sets); the thermostat the same.

4.4.3.4.5 Forward secrecy on member removal

Once shared, always shared; nothing new after removal. This is the one statement of the removal rule; CC 4.4.3.2.2 and CC 4.5.12.1 point here. A removal is never retroactive at any scope: every `key_grant` the removed member already holds stays, and no historical content is re-encrypted. After the removal the member receives nothing new, by a mechanism that differs by scope:

- **Self and family.** Every write seals under a fresh per-write DEK (CC 4.4.3.4.1), so the substrate simply stops wrapping new DEKs to the removed member's keys; no key is reused, so nothing is rotated.
- **Community and affiliations.** Content seals under the minter's epoch DEK (CC 4.4.3.2.1), which the removed member holds; after a removal each minter rotates its epoch DEK before its next sealed write, so the first write after the removal is under a key the removed member was never granted.

```
on_member_removed(scope_target, removed_member_key):
  // Existing key_grants to the removed member stay; nothing is re-encrypted.
  // self | family: stop wrapping new per-write DEKs to removed_member_key.
  // community | affiliations: each minter rotates its epoch DEK before its
  //   next sealed write for scope_target; the removed member gets no grant on it.
```

This is consistent with CC 4.5.3 "takedown isn't a coup" + CC 2.4.1 `withdraws-isn't-retroactive` semantics — historical state isn't retroactively re-keyed.

Why: matches the intuition that leaving governs future content, not historical; bounded substrate cost (one epoch rotation at most, never a re-wrap of extant content); the removed member reads nothing written after they left.

4.4.3.4.6 Self-collective resolution

For identity `I`, the current self-collective is computed as:

```
resolve_self(I, now):
  candidates = federation_attestations
                .where(subject_kind == "identity_occurrence")
                .where(envelope.identity_key_id == I)
                .where(the occurrence's current version; a version supersedes only its own earlier one)
                .where(NOT evicted <= now by a landed self-roster proposal, quorum or lone path, or by a
    node/agent single-signer revocation; CC 3.3.6.3)
                .where(envelope.valid_until IS NULL OR > now)
  return {c.envelope.occurrence_key_id for c in candidates} U {I}
```

The root `I` itself is implicitly a member (the identity_key is always an admissible signer for its own content). Single-vouch admission: any current occurrence (including `I` itself) may admit a new occurrence via `attesting_key_id` on a fresh `identity_occurrence` Contribution.

Concrete: Alice has admitted `alice_phone`, `alice_laptop`, `alice_agent`. Her self-collective is `{alice_root, alice_phone, alice_laptop, alice_agent}`. When Alice's phone publishes a `cohort_scope: self` feed scroll, the substrate wraps the content DEK under all four keys; the content reaches her laptop and agent via the at-rest encryption flow with no holding claim leaving the self cohort.

4.4.3.4.7 Composition with subject_key_ids[]

`identity_occurrence` + `family` (membership / visibility scoping) compose cleanly with `subject_key_ids[]` (revocation authority):

- A `cohort_scope: family` Contribution naming `subject_key_ids: [user_canonical_hash]` is admitted at family visibility AND the named subject retains independent revocation authority per CC 4.4.3.5.
- A `cohort_scope: self` Contribution that Alice writes about Bob (Bob in `subject_key_ids`) stays in Alice's self-collective (Bob does NOT receive a key_grant unless Bob's identity is in Alice's self — which it isn't). Bob's subject-side revocation authority still composes: Bob CAN issue `withdraws` against Alice's Contribution; if Alice committed `consent: deletion_sla`, the deletion window's clock starts at that withdrawal, and an overrun is `hard_case:deletion_window_breach` (CC 3.3.1 D2).

The orthogonality holds: **`cohort_scope` is producer-side visibility scoping**; **`identity_occurrence` + `family` are substrate-side membership primitives that gate at-rest DEK wrapping**; **`subject_key_ids` is subject-side revocation authority**. Three independent envelope-level concerns that compose without overlap.

4.4.3.5 Policy K: Consent composition

Composition pattern for dual-authority Contributions where the subject is named via CC 2.3 `subject_key_ids`, with consent state composed from the CC 3.3.1 `consent:*` namespace family.

Reads as: "this Contribution names a subject whose consent state evolves over time; consumer policy resolves the effective consent verdict by walking the subject's latest non-superseded `consent:`

`state:*` emission, gated by `valid_until`, and tracks producer deletion-SLA obligations on revocation."

4.4.3.5.1 Decay-protocol stage composition (the CIRISAgent ANONYMOUS stream)

For consent_records carrying `decay_protocol: "ciris-agent-90day"` (or any named decay path):

```
decay_state(consent_record, now):
  elapsed = now - revocation_event(consent_record).asserted_at

  walk substrate emissions on dimension `consent:decay:*` against consent_record,
  matching the decay_protocol's stage map. CIRISAgent 90-day decay:

    elapsed < 30d -> consent:decay:identity_severed (substrate emits at elapsed=0)
    30d <= elapsed < 60d -> consent:decay:patterns_anonymized (substrate emits at elapsed=30d)
    60d <= elapsed < 90d -> (in flight; no new stage emission)
    elapsed >= 90d -> consent:decay:complete (substrate emits at elapsed=90d)
```

Per CC 3.3.1 `consent:decay:{stage}` is open vocab. Other decay protocols MAY name other stage sequences; substrate honors the producer's published `decay_protocol` string and emits stages per the protocol's stage map.

4.4.3.5.2 Deletion-SLA watcher (substrate emission)

When subject `s` emits `consent:state:revoked` (or an admitted `withdraws`) against target `T`, substrate watches for producer compliance:

```
watch_sla(T, s, revocation_at):
  sla = T.attestations
          .where(attesting_key_id == T.attesting_key_id)
          .where(dimension == "consent:deletion_sla:*")
          .latest.extract_days

  if sla is None:
    return # no SLA commitment; no watcher

  deadline = revocation_at + sla.days
  completion = T.attestations
                 .where(attesting_key_id == T.attesting_key_id)
                 .where(dimension == "consent:deletion_complete")
                 .where(asserted_at > revocation_at)
                 .first

  if now > deadline and completion is None:
    emit hard_case:deletion_window_breach against T
```

The `hard_case:*` emission is the **primitive observability signal**; per CC 4.5.2 governance, a detector library composes derived detectors on top (`detection:consent:repeat_sla_breach`, etc.).

4.4.3.5.3 Bilateral pair composition (PARTNERED ceremony)

For the bilateral partnership shape per CC 3.3.5 `consent_record`:

```
ratified_pair(pair_id):
  rows   = federation_attestations
```

```
            .where(subject_kind == "consent_record")
            .where(envelope.bilateral_pair_id == pair_id)
            .where(envelope.stance == "granted")
grant  = rows.where(dimension == "consent:partnership_grant:v1").first   # signed by the user;
   subject_key_id = the agent occurrence
accept = rows.where(dimension == "consent:partnership_accept:v1").first  # signed by the agent
   occurrence; subject_key_id = the user
return grant AND accept
   AND grant.subject_key_id == accept.attesting_key_id
   AND accept.subject_key_id == grant.attesting_key_id   # both halves required; each names the other
```

`topical_relation:bilateral_pair` is the open-vocab edge documenting the pair linkage (recommended, not required for ratification — `bilateral_pair_id` is the binding mechanism).

4.4.3.5.4 Multi-subject revocation (any-subject-binding)

When `len(T.subject_key_ids) > 1`, each subject is an **independent** revocation authority. A `withdraws` admitted under CC 2.4.1.1 rule 2 or 3 from ANY single subject in `T.subject_key_ids` evicts the Contribution. Consumer policy MUST treat `T` as revoked from the perspective of all subjects (no "majority-rules" or "all-subjects-must-agree" softening) — this is the subject-as-individual principle applied at the subject-authority layer.

Concrete cases:

- Group photo with three subjects: any one subject revokes → the photo is evicted from federation propagation.
- Group chat export with N participants: any one participant revokes → the export is evicted.
- Multi-party contract: any one signatory revokes → the contract Contribution is evicted (separate from the legal-validity question, which is consumer-side; the substrate just removes the wire artifact).

Producers MAY mitigate by partitioning content into per-subject Contributions (e.g., one chat-message Contribution per author, linked via `topical_relation:replies_to`) so that one subject's revocation doesn't evict another's content.

> ***No distinct multi-subject evict path — admission + precedence is sufficient.*** *Multi-subject eviction is fully expressed by two mechanisms already in the spec and needs* ***no new primitive, dimension, or*** `hard_case`*: (1)* ***admission*** *— the per-subject* `withdraws` *is admitted under CC 2.4.1.1 rule 2/3 exactly as a single-subject* `withdraws` *is (each subject in* `subject_key_ids` *is independently a valid revoker; the rule does not change for* `len > 1`*); (2)* ***precedence*** *— the latest-wins / revoke-is-terminal precedence at the consumer read path (CC 4.4.3.5.5) applies per-subject, and any one subject's terminal* `withdraws` *evicts* `T` *for all (the any-subject-binding above). There is no quorum to compute and no "evict event" to materialize separately from the admitted* `withdraws` *— the withdrawal IS the evict. Substrates implement this as the OR over per-subject revocation state, not as a new code path.*

4.4.3.5.5 Effective consent resolution (read path)

For a target Contribution `T` carrying `subject_key_ids` of length ≥ 1, the effective consent state per subject `s` $\in$ `T.subject_key_ids` is computed as:

```
resolve_consent(T, s, now):
  rows = federation_attestations.where(target == T).where(attesting_key_id == s OR
    attesting_key_id in delegates_to(s).proxies).where(dimension matches "consent:state:*")
  if rows is empty: return unspecified      // subject named but never declared

  // First, an admitted `withdraws` has precedence over every state row: a
  // withdrawn grant resolves to unspecified, and contributes no state below
  withdrawn(row) = exists w in admitted withdraws: w.target == row.id
  rows = rows.where(NOT withdrawn(row))
  if rows is empty: return unspecified      // every declaration withdrawn

  // Second, a revoke is terminal for the grant it names (its supersedes_id); a revoke that
  // names no grant revokes every grant from s asserted at or before it. A later
  // grant is a new grant with a new id and is resolved on its own: no replay of a
  // revoked grant, and no later-dated copy of it, reopens it.
  revokes = rows.where(dimension == "consent:state:revoked")
  revoked(g) = exists r in revokes: r.supersedes_id == g.id OR
               (r.supersedes_id IS NULL AND r.asserted_at >= g.asserted_at)
  // Third, latest-wins among the grants that remain
  grants = rows.where(dimension == "consent:state:granted").where(NOT revoked(g))
               .where(NOT superseded by a later grant in its supersedes chain)
               .order_by(asserted_at DESC)

  g = first(grants)
  return:
    revoked if g is null AND revokes is not empty
    expired if g is null                      // only expired or superseded rows remain
    expired if exists row "consent:state:expired" with supersedes_id == g.id OR
      g.valid_until passed without renewal
    granted otherwise
```

Substrate MAY cache the resolution per `(T, s)` keyed on the latest `asserted_at`; invalidate on any new `consent:*` write from `s` or `s`'s proxy chain.

4.4.3.5.6 What Policy K composes

CIRISAgent consent stream	Policy K composition
TEMPORARY (14d default)	`consent:state:granted` + envelope `valid_until = asserted_at + 14d` + auto-renew dimension on interaction (consumer-policy concern)
PARTNERED	Bilateral pair per CC 4.4.3.5.3: subject `consent:partnership_grant` + producer `consent:partnership_accept` under same `bilateral_pair_id`; no `valid_until`
ANONYMOUS	Revocation + decay-protocol per CC 4.4.3.5.1: substrate emits stage milestones; agent honors stage-appropriate processing constraints

CIRISAgent's three streams are a **named bundle** at the consumer-policy layer. Other agents MAY compose other streams over the same wire primitives. CEG documents the canonical bundle for ecosystem coordination; CEG does not lock the bundle.

4.4.3.6 Policy I: Attestation-Ladder Composition

The familiar L1-L5 verification "ladder" (self_verify → hardware_rooted → registry_consensus → license_validity → agent_integrity) is **consumer-side composition over the mechanism prefixes** in CC 3.1.2, not a wire-level taxonomy.

Per CC 1.2 T2 honestly applied: the L-number names a *ladder position* (a verdict-shape consumers compute), not a *mechanism* (which is what the wire MUST carry). Prefixes like `attestation:l3:registry_consensus` smuggle the verdict-shape into the prefix name, conflating the mechanism (registry consensus check) with the ladder slot (third rung). The wire separates them.

Wire prefixes (mechanism) CC 3.1.2:

Mechanism prefix	Ladder position (consumer-rendered)
`attestation:self_verify`	L1
`attestation:hardware_rooted`	L2
`attestation:registry_consensus`	L3
`attestation:license_validity`	L4
`attestation:agent_integrity`	L5

Composition function (reference implementation):

```
ladder_verdict(attestations) =
  let levels = []
  for prefix in [self_verify, hardware_rooted, registry_consensus,
      license_validity, agent_integrity]:
    if any positive attestation on attestation:{prefix}:
      levels.push(prefix_to_ladder_position(prefix))
  return {
    achieved: max(levels) if levels else None,
    ladder: sorted(levels),
    rendering: format_as_l1_l5_for_ui(levels)
  }
```

Consumers MAY render the ladder as `L1` / `L2` / `L3` / `L4` / `L5` for UI / dashboards / audit trails. The rendering is composition output, not wire emission.

Why this matters: a verifier emitting `attestation:registry_consensus +1.0` is the mechanism claim. Whether that's "L3" in any particular consumer's ladder ordering is a composition concern — different consumers may order or weight the rungs differently (e.g., some safety-critical applications may require L4 *and* L5, others may treat L3 as sufficient for advisory work). The wire stays neutral; the ladder is consumer policy.

Mechanism-only emission: emissions use the `attestation:{mechanism}` form per the table above. The `attestation:l{N}:*` form is rejected at admission per the CC 1.2 gate.

4.4.3.7 Policy F: `agent_files` trust composition

Three-layer consumer policy for composing trust over `agent_files:*` attestations (CC 3.1.9.1 + CC 3.1.1).

Layer 1 — Default trust: the shipped default root's conferring roster publishes the default trust set; an `agent_files:*` attestation at or above the root's published threshold from a key of that roster puts a file in the default-trust state; the shipped default root publishes `score` $\geq$ `0.7`. It is the shipped default, never the root (CC 4.4.3.8 Policy A): a consumer that pins another root takes that root's published set. The default root's install endpoint resolves default files via this rule.

Layer 2 — Open Contribution: any federation-key holder may emit `agent_files:*` attestations. The wire format admits them; consumer policy decides whether to surface them. The "Browse alternatives" view shows third-party agent_files with explicit provenance disclosure.

Layer 3 — Vote-then-trust: a non-canonical `agent_files:*` attestation accumulates votes. Consumer policy may elevate trust once an accumulated-weight threshold is reached.

Anti-tricking guarantee: the Layer 1 rule applies at the install endpoint **regardless of attester or vote accumulation**. Third-party agent_files are reachable only via the explicit "Browse alternatives" path. This binds the default root's own roster as it binds anyone: a newcomer's default trust path is the published default set, and nothing but the choice of another root changes it.

4.4.3.8 Policy A: direct trust

Consumer trusts an attestation if `attesting_key_id` is in the consumer's pinned trust set (canonical bootstraps + consumer-added pins). Cheapest, lowest-latency, narrowest reach.

Aggregation: per (`dimension`, `attested_key_id`) tuple, over trusted attesters, by the polarity defaults of CC 4.4.2: the mean of `score` × `confidence` applies only where the dimension's polarity is `signed`; a boolean-via-score dimension takes the minimum, a positive-only dimension the maximum, a detector dimension the median. Consumer threshold determines verdict.

Recommended default: Policy A with `pinned_trust = {community_key_id: ciris-canonical, family: humanity-accord}` — the founders of `ciris-canonical`, who are the three accord holders `A1`, `B1`, `C1` (CC 3.2), resolved live, never a serving install's key. Cold-start bootstrap: a new consumer fetches `GET /v1/trust-root/bundle` (CC 5.3.4), re-derives the bundle's quorum from the holder records it carries against out-of-band pins (the published holder fingerprints, or a build-baked copy of the bundle), holds a witnessed lineage head inside the root's `attach_window` (CC 3.2 T4a; served beside the bundle per CC 5.3.4), and persists the anchors locally; `GET /v1/accord-holders` is a projection it cross-checks, not a second root. This is the shipped **default** root — never *the* root; the model below is what makes that distinction structural rather than rhetorical.

The pinned trust set is a graph, and its roots are pluggable (normative).

1. **Self-root base.** First run writes a plain `attestation(user → user)` — "I trust my own attestations and service" — with no modifiers. This is the immutable identity floor: the user is their own root. A node inherits it by owner-binding `delegates_to(user → node)` (CC 3.2); an agent by the partnership object (CC 4.4.3.4.3).
2. **A root is a self-referential charter.** `delegates_to(root → root, scopes)` — rooting to itself *is* rootness. Because down-chain grants are **attenuation-bound** (a grant resolves only where every edge on the chain bears the scope; there is no amplification), that self-declaration is the **charter**: the capability ceiling of the entire trust domain, held in signed bytes and re-resolved by live graph walk, so an amendment re-scopes existing grants with no re-issuance. **Root validity minimum `[infra:serve, infra:attest]`** — a root serves and vouches, or it is inert. The humanity accord's charter is `[infra:attest, infra:serve, infra:store, infra:transport]`.
3. **Roots are pluggable; validity flows from a shared root.** A consumer trusts one by hanging the acceptance edge off the base: `delegates_to(user → root)` bearing `trust:accepts:v1` (CC 3.2 T3), added and removed freely, base untouched. Two nodes under one shared root cross-attest and vouch; two nodes with no shared root compose nothing. The sanctioned bootstrap and re-root artifact is a **portable trust root**: a charter plus at least one blessed `infra:serve` node, minted by the CC 4.2.1 conferral ceremony and owing

nothing to any prior root.

4. **Un-trust is nuclear but recoverable.** Deleting the `user → root` edge revokes, at once, that root's conferred roles, score validity for this consumer, vouching, **and its kill-switch authority** (CC 4.2.1). The base self-root survives, so the installation keeps running unfederated; attaching a new root resumes federation. Un-trust is the exit that makes the halt consensual.
5. **Capability gate — the federation floor (fail-closed).** With no valid *external* trust root, federated agent capabilities and manifest validation MUST be disabled at the server tier. The base self-root is the floor, never sufficient for federation: it constitutes an identity, and identity is not standing (CC 4.1.2).
6. **Two conferral planes. Ceremony** confers root *identity* (an m-of-n co-scrub on the key record, CC 4.2.1); **delegation** confers operational *capability* (`delegates_to(root → key, [scope])`, resolved at use against a root the asking node trusts). Every reserved-prefix role — `trusted_publisher` reserving `content_rating:*` (CC 4.4.3.10), `lenscore_detector` reserving `detection:*` — is conferred on the **delegation** plane. Putting a routine operator role on the ceremony plane fails closed on every legitimate operator and leaves each newly-minted root *less* able to stand up its own; leaving a reserved-prefix role with **no** named plane leaves it self-assertable, which is the hole this rule closes.
7. **Charter recovery lives in the record.** A charter MUST carry a pre-rotation commitment (the successor key committed in advance) and an m-of-n recovery ceremony, and the m-of-n MUST be audited for holder **independence** — an m-of-n assembled inside one organisation is a 1-of-1. Absent both, root-key compromise is unrecoverable by construction: the attacker holds the tombstoning pen.
8. **Genesis is tamper-evident, not self-verifying.** A portable-root bundle verifies internally — and so does an attacker's identical-shaped bundle. The trust-on-first-use moment therefore sits in the distribution channel, so out-of-band fingerprint comparison is a **first-class attach step**, not an optional hardening.

1+4 preserved — every element above composes the existing `attestation` and `delegates_to` primitives; no new structural primitive, no new envelope field, and no bespoke "trust" attestation kind.

4.4.3.9 Policy D: Lexical-vulnerability-priority

A composition tie-breaking rule layered on top of any base policy. When two otherwise-equivalent attestations conflict, defer to whichever attestation names the more-affected cohort — measured by `affected_population_estimate` in the attestation `context`, weighted inversely (smaller = more vulnerable, more weight).

Inverts the default popularity-weighted aggregation specifically for ties. Consumer policy, NOT a wire-format primitive.

4.4.3.10 Policy J: Trusted-Publisher composition

Composition pattern for multimedia content discovery. Reads as: "this `external_content` Contribution comes from a publisher whose attestation chain is trusted at the cohort level, with content-class + content-rating + age-assurance composed into the gate."

The composition has three layers (analogous to CC 4.4.3.7 Policy F for agent_files but specialized for multimedia content):

Layer 1 — Distributor attestation chain: an `external_content` Contribution with `sub_kind: film` (or any media sub_kind) carries a distributor attestation that chains to a federation_key with `identity_type: distributor`. Distributor identity is established via CC 3.1.1 Registry partner_role machinery + an out-of-band distributor-onboarding flow (operator's choice — the shipped default root's conferring roster publishes the default trust set; a substrate that pins another root maintains its own).

Layer 2 — Content-class + content-rating composition: consumer gates by combining `content_class:{class}` + `content_rating:{scheme}:{rating}` per CC 3.3.12:

```
gate_decision(content) = match (content.content_class, content.content_rating, consumer.preferences):
  # Producer-declared content_class is consultable but not authoritative -- UI may show
  # the producer claim alongside cohort cw_class declarations and let the consumer choose.
  (class, _, prefs) if class in prefs.blocked_classes => Block
  (_, rating, prefs) if rating.exceeds(prefs.max_rating) => Block
  (_, _, _) => Allow
```

Layered with CC 4.4.1 — `cw_class:*` declarations from low-attestation-density cohorts MUST NOT be downweighted; they ride alongside the gate decision as informational.

Layer 3 — Age-assurance gating: for content where `content_rating:*` rises above an age threshold (e.g. PEGI 18, MPAA NC-17), consumer gates via `age_assurance:{level}` per CC 3.3.12:

```
age_gate(content, consumer):
  required_level = age_required_for(content.content_rating)
  consumer_level = consumer.highest_age_assurance_level
  return consumer_level >= required_level
```

Where the `age_assurance:{level}` ordering is: `self < provider:{verifier_key}:adult < government:{credential_class}:adult`. Consumer SHOULD accept the strongest assurance the user has provided; substrate MUST NOT issue `slashing:*` on age-assurance misdeclaration alone — `moderation:age_assurance_misdeclaration` is the adjudication path per CC 3.1.9.2.

Anti-tricking guarantee parallel to CC 4.4.3.7: the canonical-distributor Layer 1 rule MUST apply regardless of vote accumulation. No amount of accumulated vote weight elevates an unverified distributor into Layer 1; the only path is the operator-set trust list. This binds the default root's own roster as it binds anyone: it cannot exempt its own content distribution from this rule.

4.4.3.11 Policy G: Trust-Fresh / Lighthouse

Composition pattern: `cert_validity:{authority} + transparency_log:inclusion + (attestation:regi` `OR attestation:license_validity)`, the "freshness + attested + verified" idiom.

Reads as: the consumer wants confirmation that (a) the cert chain is currently valid; (b) the attestation appears in a transparency log; (c) either `attestation:registry_consensus` (the ladder's L3 position per CC 4.4.3.6 Policy I) or `attestation:license_validity` (L4) is satisfied. The combination is the consumer-side recipe for "this attestation is fresh AND verified AND multi-source-consensus-backed."

Not a wire primitive; a recognized composition pattern that consumer libraries SHOULD expose as a named one-call helper.

4.4.3.12 Policy B: one-hop transitive

Consumer trusts an attestation if `attesting_key_id` has been vouched for by the pinned trust set. Adds one hop of indirection.

4.4.3.13 Policy C: weighted graph

Consumer applies transitive-trust propagation across the full attestation graph, weighted by canonical-bootstrap distance with confidence decay per hop. Requires more compute; less common in practice; needed for federated reputation across many partner orgs.

4.4.4 Sovereign-Registered equivalence (wire-symmetric, policy-differentiated)

A Sovereign agent scoring `licensure:CA_medical_board: +1.0` is wire-format identical to a Registry-steward scoring the same. Consumer policy weights by attester source; the substrate is source-neutral.

Both paths produce federation membership; neither is a gate. What differs is the *attestation surface* — the kind of claim the federation can compose about why a participant is trustworthy.